

BEGINNER'S GUIDE: YOUR FIRST AUTONOMOUS AI IN SAPPHIRE

Welcome! This guide teaches Sapphire from scratch. By the end you will have a working autonomous AI bot and understand how Sapphire frontier LLM training works.

Step 1: Hello World

```
print("Hello from Sapphire!");
```

Step 2: Variables and Functions

```
let name = "Sapphire"; let version = 1;
fn welcome(lang) { print("Welcome to {lang} v{version}!"); }
welcome(name);
```

Step 3: Your First AI Query

```
let question = "What is the capital of France?";
let answer = ai.prompt(question);
print("AI says: {answer}");
```

Step 4: PC Automation

```
let info = os.system_info();
print("RAM used: {info.ram_percent}%");
print("CPU: {info.cpu_percent}%");
os.notify("Sapphire Bot", "System check complete!");
```

Step 5: Your First Autonomous Bot

```
fn health_bot() {
  let stats = os.system_info();
  let advice = ai.prompt("RAM is {stats.ram_percent}%. What to do?");
  print("Bot advice: {advice}");
  os.notify("Health Bot", advice);
}
health_bot();
```

Step 6: Concurrency Made Easy

```
parallel {
  { print("Task 1: checking RAM..."); }
  { print("Task 2: fetching news..."); }
  { print("Task 3: writing log..."); }
}
```

Step 7: Introduction to Frontier LLM Training

Sapphire can train LLMs at frontier scale (70B-540B parameters, 512 GPUs) using ml.distributed:

```
let model = ml.distributed.Transformer({"layers": 32, "hidden": 4096,
  "heads": 32, "vocab": 32000, "precision": "bf16"});
let cluster = ml.distributed.Cluster({"gpu_type": "H100-80GB", "num_gpus": 8});
let result = ml.distributed.train(model, cluster,
  {"tokens": 10000000000, "batch_size": 512});
print("Done! Strategy: {result.strategy}");
```

KEY: The 5D Auto-Parallelism Solver automatically picks the best training strategy.

You do not need to manually tune TP/PP/DP/EP/SP -- Sapphire does it for you!

Quick Reference Card

SAPPHIRE v1.0.0 | Beginner's Guide: Your First Autonomous AI

Action	Sapphire Code
Print	print("Hello {name}");
Variable	let x = 42;
Function	fn my_fn(a) { print(a); }
AI Query	let ans = ai.prompt("question");
System Info	let s = os.system_info();
Parallel	parallel { { task1(); } { task2(); } }
Frontier Train	ml.distributed.train(model, cluster, cfg);
Notify	os.notify("Title", "Message");